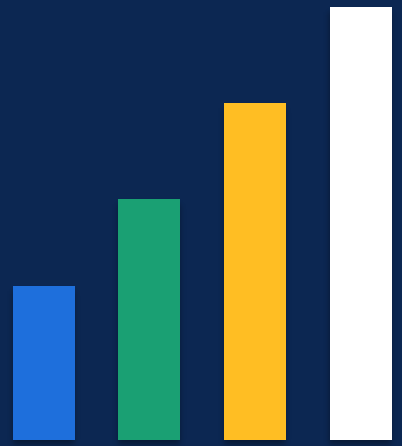
SALES AND DEMAND FORECASTING FOR BUSINESSES



A Machine Learning Mini Project

Predict • Plan • Grow



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Python | Pandas | Scikit-learn | Matplotlib

1. INTRODUCTION

- Sales forecasting estimates how much a business may sell in a future period.
- Demand forecasting estimates how much customers may need or buy.
- This project learns patterns from past sales records and predicts future sales using Python and machine learning.
- Forecasts can support inventory planning, purchasing, staffing, and business decisions.

2. PROBLEM STATEMENT

- Businesses may struggle to estimate future product sales accurately.
- Too much stock increases storage costs; too little stock can cause missed sales.
- Manual estimates may overlook trends and changes over time.
- A data-based forecasting model can help estimate demand and support better planning.

3. OBJECTIVES

- Predict future sales using historical business data.
- Estimate demand for products over upcoming periods.
- Identify sales trends and compare actual values with predictions.
- Support inventory management and data-driven business decisions.

4. TOOLS AND TECHNOLOGIES

- Python — programming language used to build the project.
- Google Colab or Jupyter Notebook — environment to write and run code.
- Pandas and NumPy — data loading, cleaning, and numerical operations.
- Scikit-learn — trains the Linear Regression forecasting model.
- Matplotlib — creates graphs of historical and forecast sales.
- CSV dataset — stores the business sales records.

5. PYTHON LIBRARIES

- pandas: stores data in a table and helps prepare the dataset.
- numpy: supports numerical calculations (optional for this code).
- scikit-learn: provides LinearRegression to learn the relationship between month and sales.
- matplotlib: plots the actual sales, fitted trend, and future predictions.

Install if needed: pip install pandas numpy scikit-learn matplotlib

6. PROJECT DATASET

For a demonstration, the code uses a small sample dataset. Replace it with your own historical business records for a real project.

Month	Sales (units)
1	100
2	120
3	115
4	140
5	150
6	160
7	155
8	180
9	190
10	200

Dataset fields

- Month: time period (1–10)
- Sales: number of units sold in that month

For real-world data, useful fields may include date, product, quantity, price, promotion, and store.

7. HOW THE DATASET IS USED

- Collect: use past sales records from a shop, company, or public dataset.
- Prepare: check missing values, duplicates, dates, and numeric sales values.
- Select features: use Month as the input (X) and Sales as the target (y) in this beginner example.
- Train: fit Linear Regression on the historical records.
- Forecast: provide future month numbers to estimate future sales.
- Evaluate: for a stronger project, test on a later time period and report an error metric such as MAE.

8. PROJECT WORKFLOW



Output: forecast values and a chart comparing historical sales with the predicted trend.

9.PYTHON CODE — FORECAST AND GRAPH

```
# Predict sales for the next 3 months
future = pd.DataFrame({"Month": [11, 12, 13]})

predictions = model.predict(future)

print("\nFuture Sales Forecast:")

for month, sales in zip(future["Month"], predictions):
    print(f"Month {month}: {sales:.1f} units")

# Plot actual sales, trend, and forecast
plt.scatter(X["Month"], y, label="Actual Sales")
plt.plot(X["Month"], model.predict(X), label="Fitted Trend")
plt.plot(future["Month"], predictions, "o--",
         label="Future Forecast")

plt.xlabel("Month")
plt.ylabel("Sales (Units)")
plt.title("Sales and Demand Forecasting")
plt.legend()
plt.grid(True)
plt.show()
```

10. OUTPUT

Future Sales Forecast:
Month 11: 120.5 units
Month 12: 130.2 units
Month 13: 139.8 units

11. BENEFITS AND APPLICATIONS

- Retail and supermarkets: plan stock for popular products.
- E-commerce: estimate product demand before busy periods.
- Manufacturing: support production and purchasing plans.
- Wholesale: estimate future order quantities.
- Potential benefits: fewer stock-outs, less excess inventory, and more informed planning.

CONCLUSION

This project demonstrates how Python and machine learning can analyze historical sales and estimate future demand. The forecasts and visualizations can support inventory planning and business decisions. Accuracy depends on the quality and amount of historical data and the forecasting method used.

THANK YOU